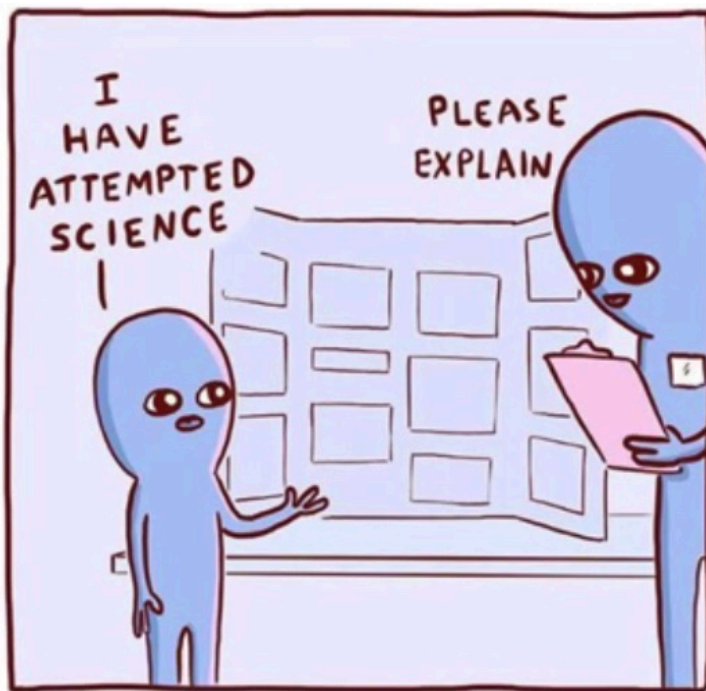


How do scientists study development and learning?



NATHANWPYLE

T/Th 9:35-10:25am

Professor Casey Lew-Williams

Kennedy Casey, Abby Fergus, Sarah Joo, Chantal Valdivia, Claire Whiting

“The amount of energy needed to refute bullshit is an order of magnitude bigger than that needed to produce it.”

Alberto Brandolini, 2013

“Falsehood flies, and truth comes limping after it.”

Jonathan Swift, 1710

Outline for today

- Ethical considerations in developmental research
- Open Science and Pre-Registration
- W.E.I.R.D. Participants
- Types of research studies
 - Correlational, experimental, descriptive
 - Longitudinal vs. cross-sectional studies
- Methods in developmental science
 - ‘Traditional’ developmental methods
 - Examples of specific research methods/tools

Ethical considerations

- Ethics for child and adult participants
- Researchers have a vital responsibility to
 - anticipate potential risks that children in their studies may encounter,
 - minimize any risks,
 - and make sure that the benefits outweigh any potential harm.



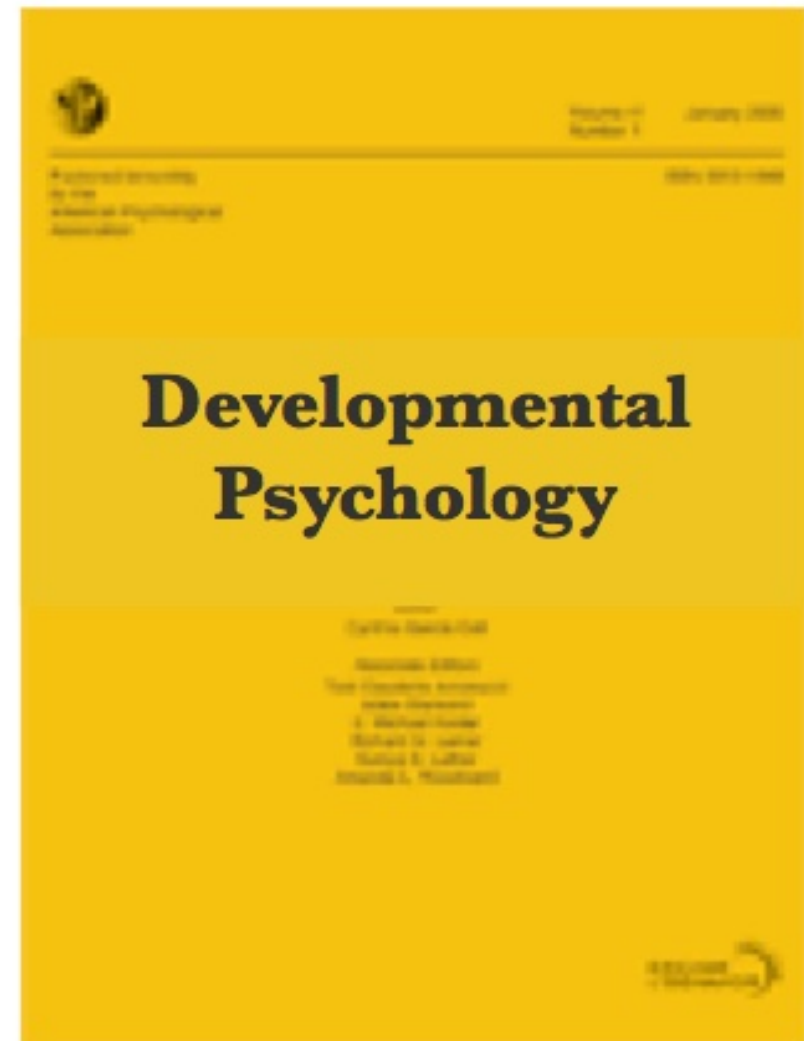
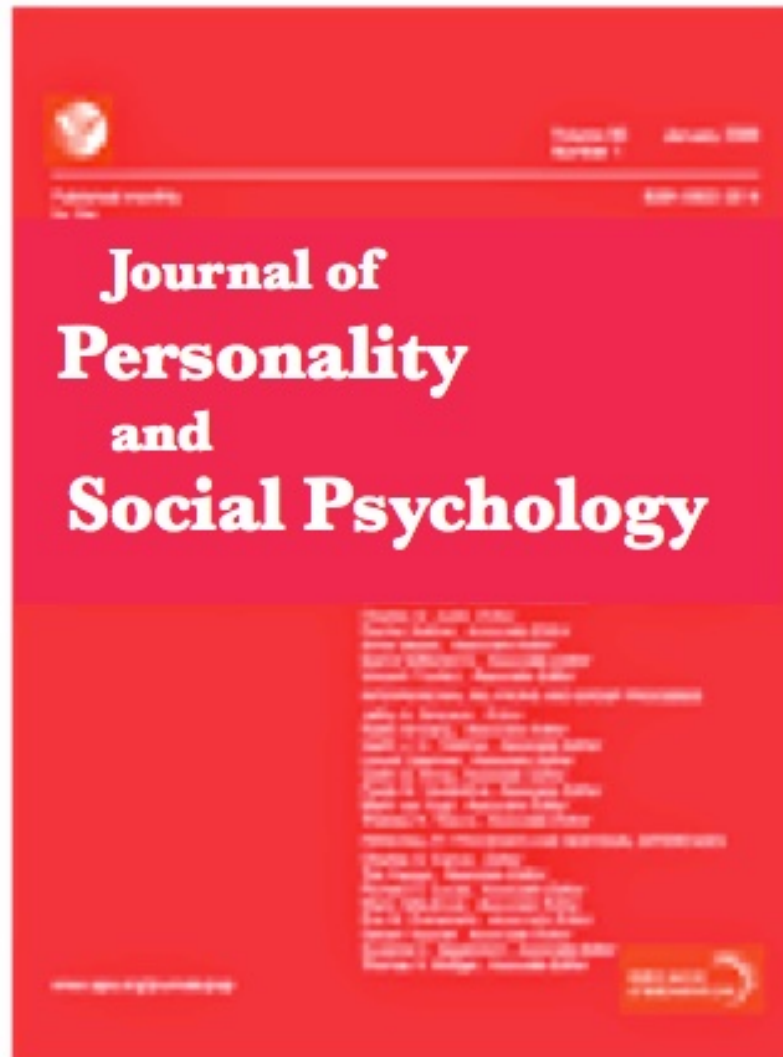
Open Science and Pre-Registration

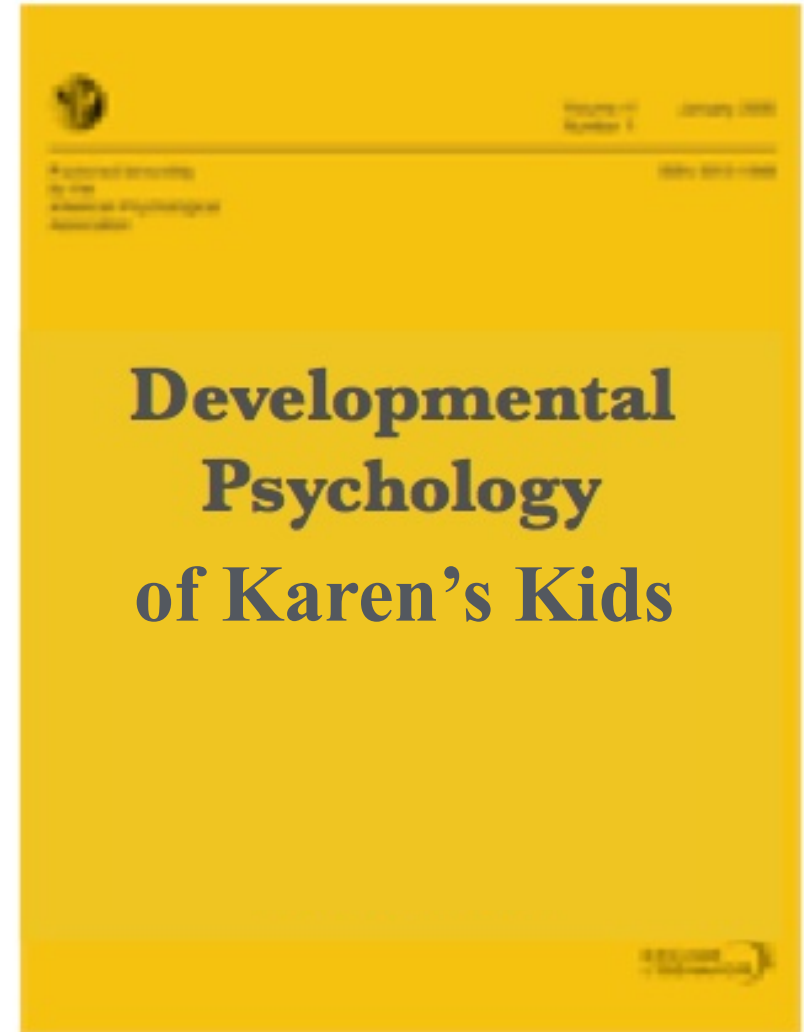
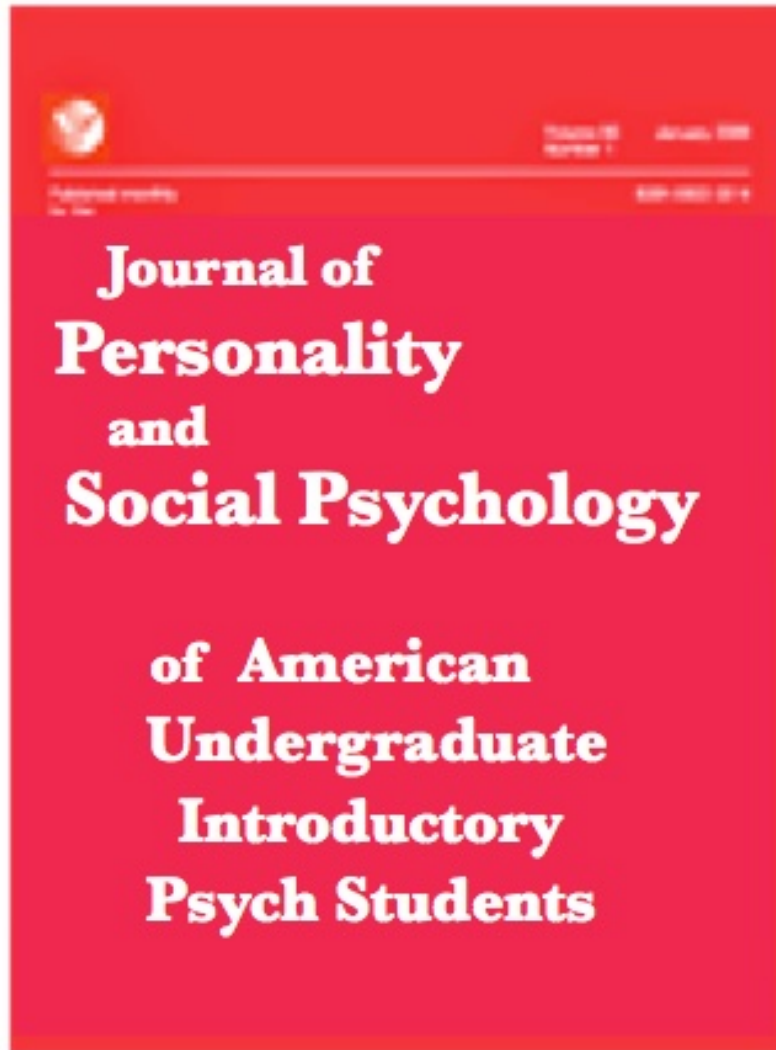
Open science: making science accessible to anyone from beginning to end.



W.E.I.R.D. participants

- **W**estern, **E**ducated, and from **I**ndustrialized, **R**ich, and **D**emocratic countries.
- 12% of countries fall in this category
- 68% of psychology studies are from the United States.
96% are from North America, Europe, Australia, Israel.
- 67% of U.S. participants are psychology students
- Also, most experimenters are W.E.I.R.D. too

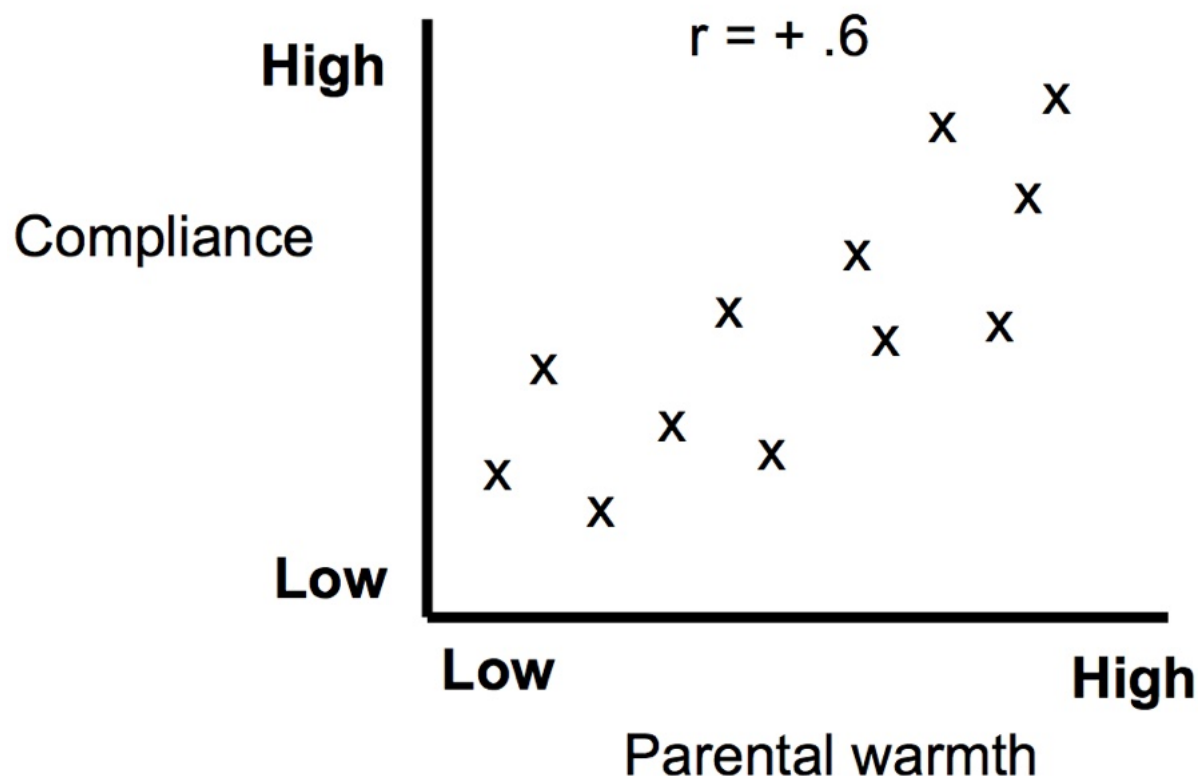




Types of research studies

Correlational studies

- Correlational studies
 - Measure a set of variables, see if they're related
 - Example: children's compliance and parental warmth



Correlational studies

- **Kid shoe size is correlated with reading skill.**
 - Does shoe size cause reading skills?
 - What's the latent variable?
- **Everybody who eats carrots dies.**
 - Are carrots lethal?
- **Sleeping with a night light is related to later nearsightedness.**
 - Conclusion: night lights cause vision problems.
 - Alternative explanation?

Correlational studies

eating and health

Whole Milk Or Skim? Study Links Fattier Milk To Slimmer Kids

by ALLISON AUBREY

March 20, 2013 12:27 PM ET

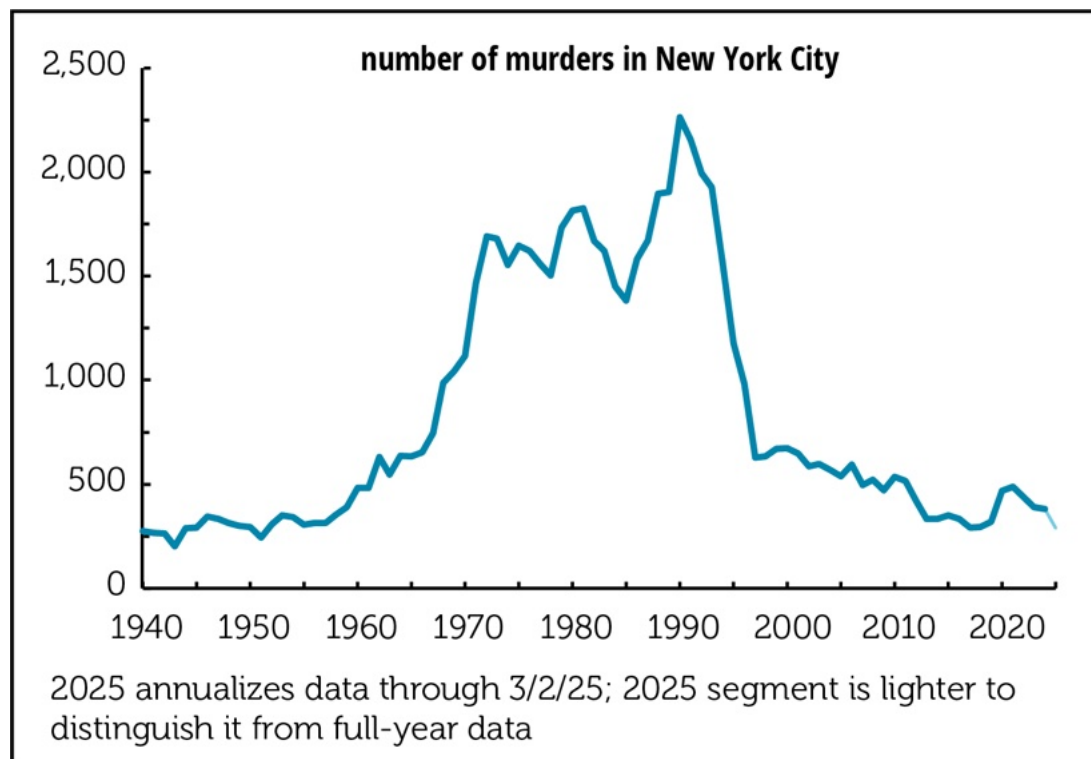


Share

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THE
BULLY
DORIS KEARNS



mountains in
the Dolomites

Experimental studies

- **Independent variable (IV)**
 - Manipulated by experimenter
- **Dependent variable (DV)**
 - Behavior measured by experimenter
 - Examples: selection of objects, ratings from 1-7, reaction time, listening/reading times, response to questions, physiological or neural responses...
- Hold other factors constant
- Random assignment vs. non-random assignment

Example: Mean Monkey study



- Research question?
- IV?
- DV?
- Is this a good design?

Descriptive studies

- Psychology has long been focused on the idea of studying individuals. But, it turns out this can only get you so far.
- Psychology should make more room for careful descriptions of behavior *in context*.
- Why are scientists so obsessed with controlled, lab-based experiments? Why not embrace the raw complexities of the environment?



Longitudinal studies

- Follow same children over time
- Strengths?
- Weaknesses?



Cross-sectional studies

- Test different children at different ages
- Strengths?
- Weaknesses?

Let's practice

How would you design a study
about the most effective ways
to get kids to comply?

Methods in Developmental Psychology

APGAR: the first and most important test you ever took!

Newborns get 0, 1, or 2 points for each of five categories:

Apppearance (blueish; pink body; pink body *and* extremities)

Pulse (0; <100; >100 bpm)

Grimace/reflex (none; slight response; cry or pull away)

Activity (none; some; flexed)

Respiration (absent; weak; strong)



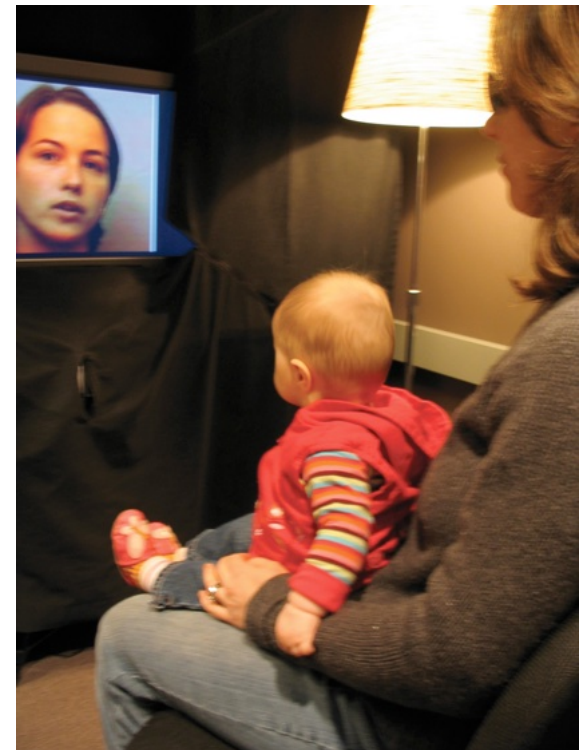
'Traditional' developmental methods

- Diary studies (e.g., Charles Darwin in the 1850s; Marilyn Shatz in the 1980s)
- Interviews
- Questionnaires/surveys

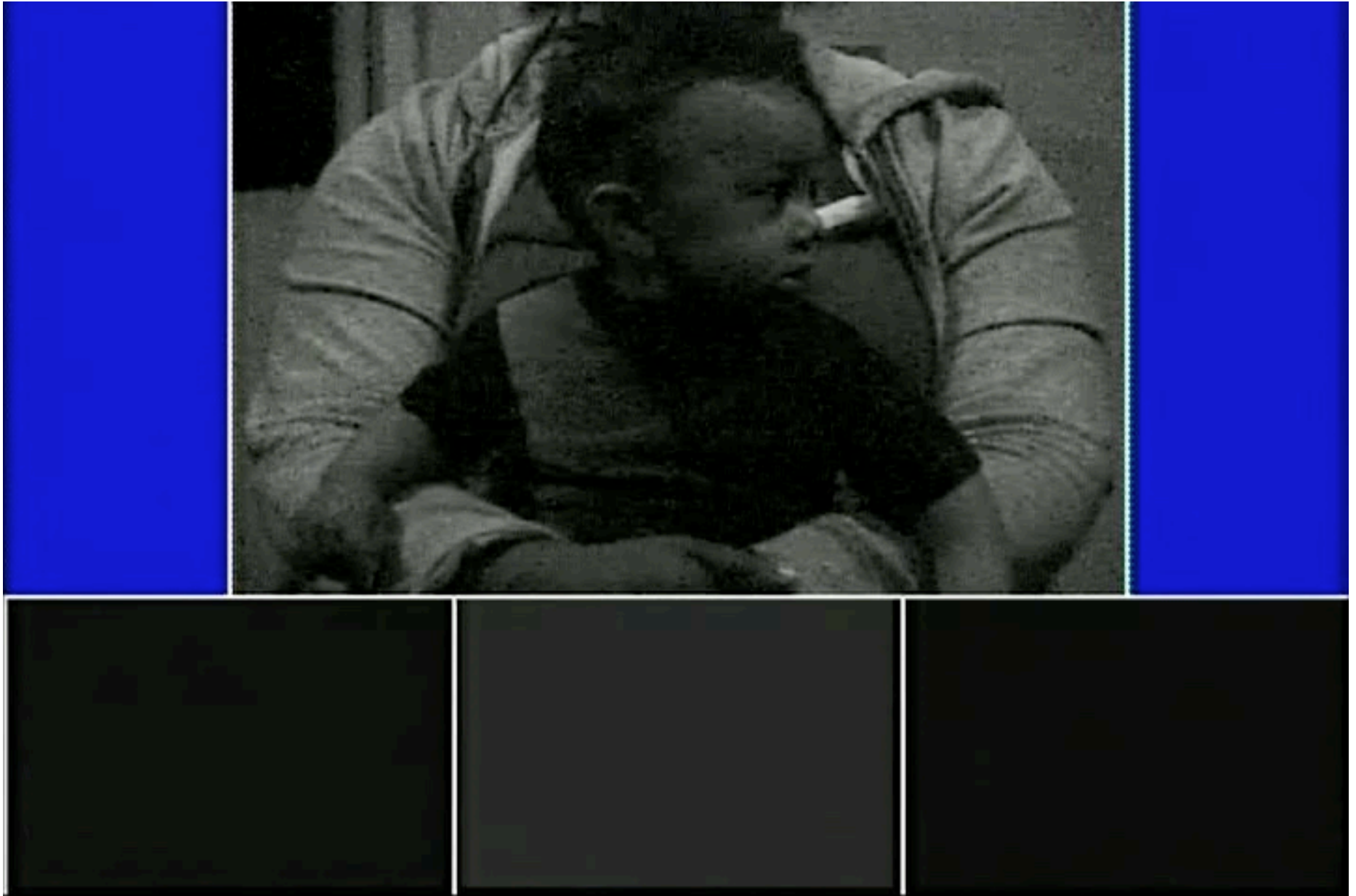


Habituation

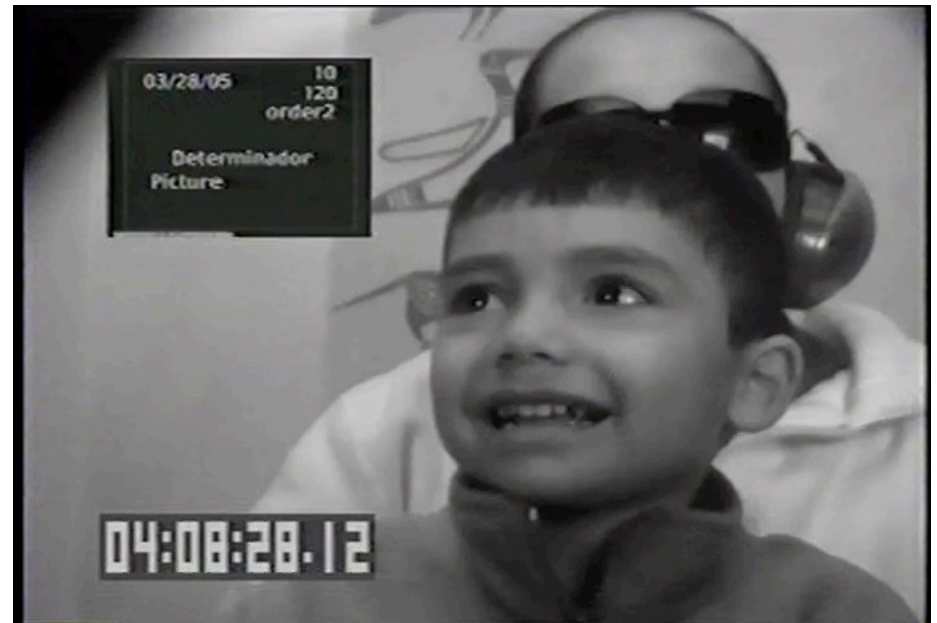
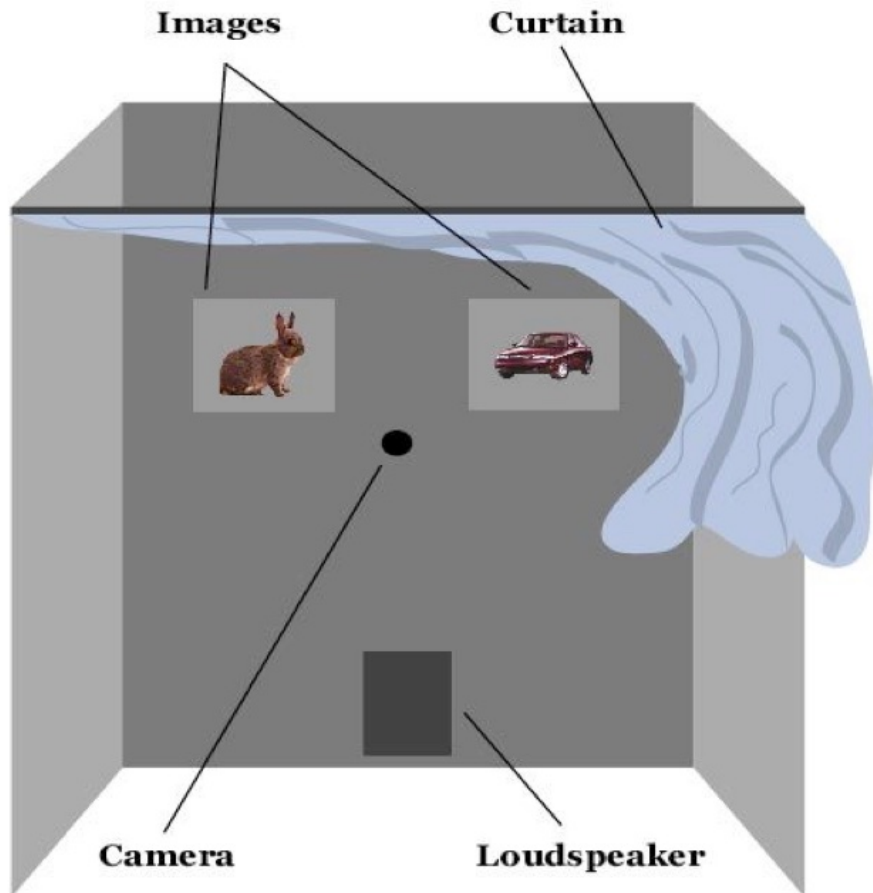
- Show something to babies over and over until they 'habituate' to it (i.e., until their looking time declines to half of what it was compared to the initial time)
- Then, show them something new and see if they look longer



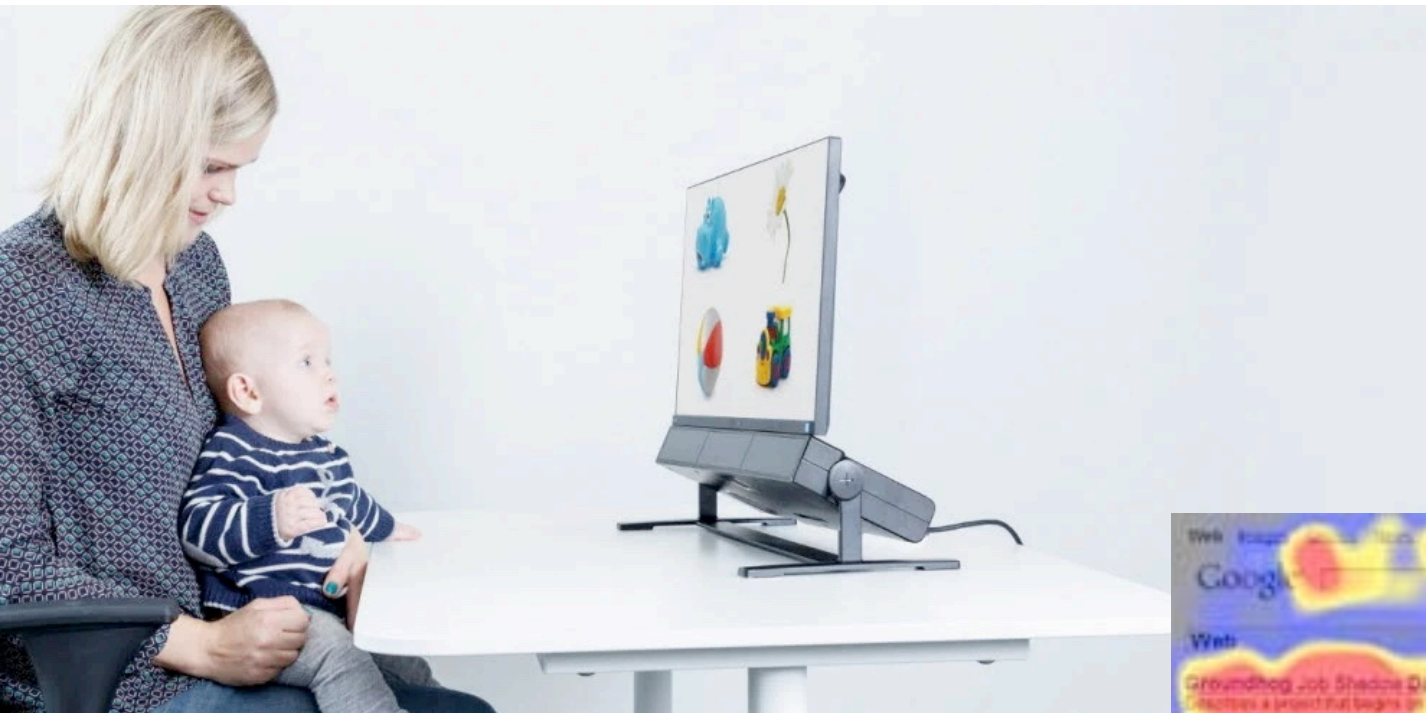
Headturn preference procedure



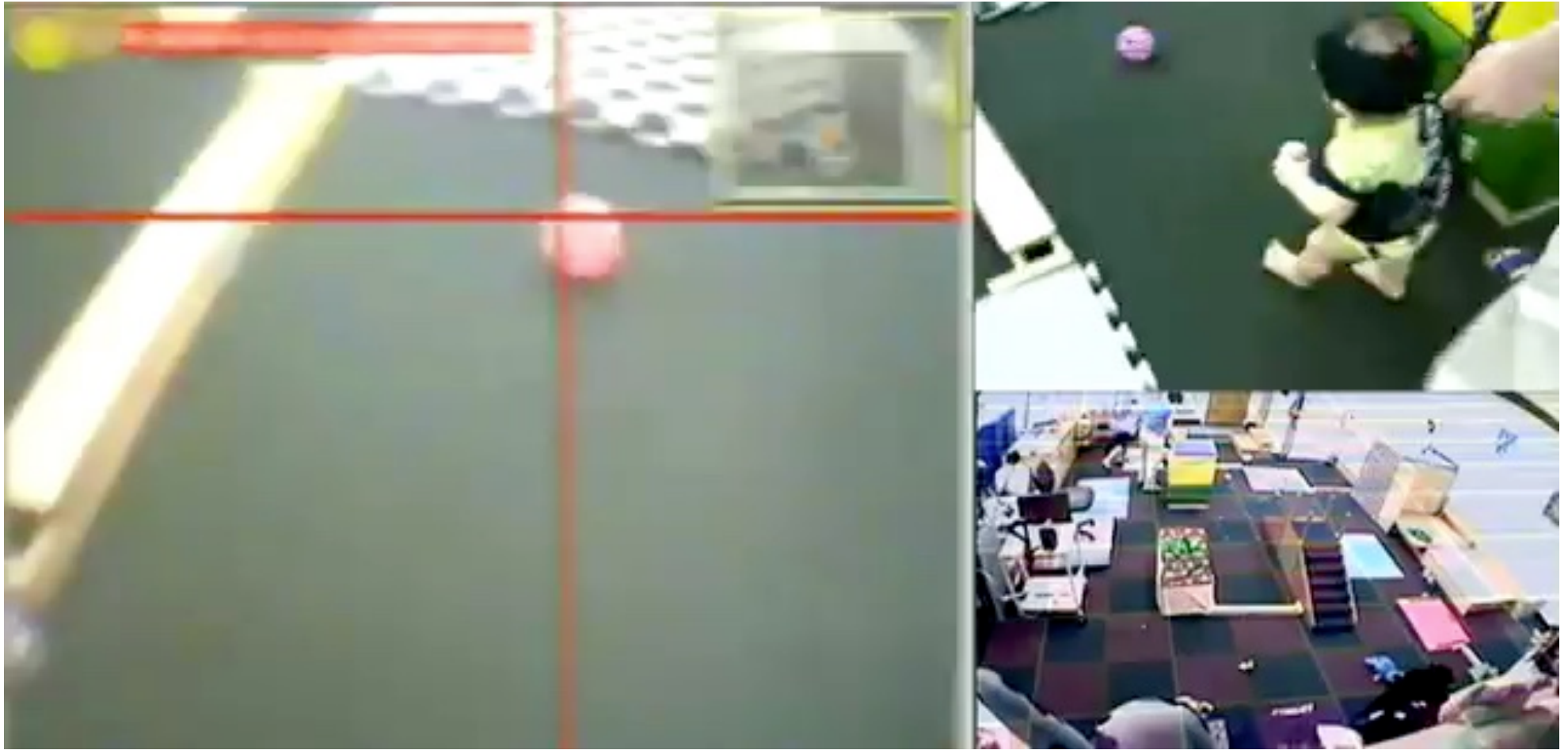
Looking-while-listening procedure



Eye tracking



Eye tracking in action



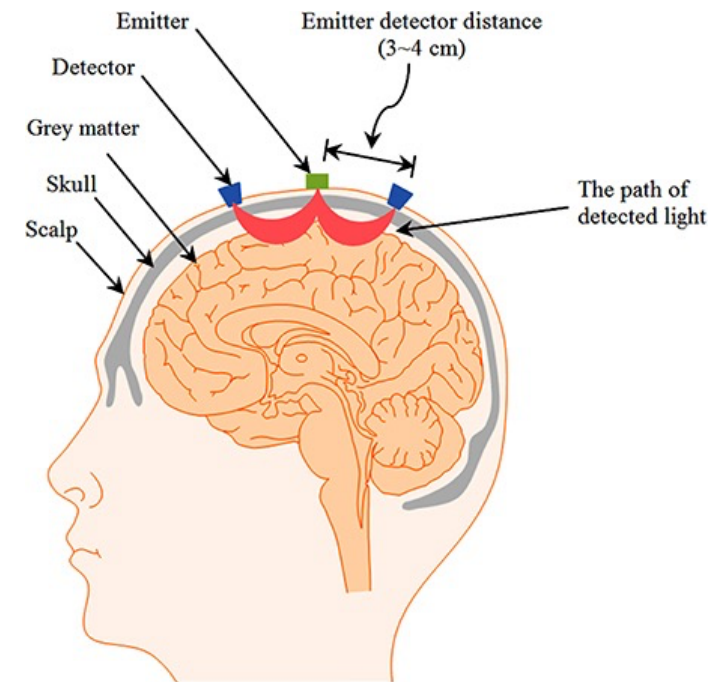
Physiology: pupil size, heart rate, cortisol



Cognitive / emotional processing



fNIRS: functional near-infrared spectroscopy (brain imaging using near-infrared light)



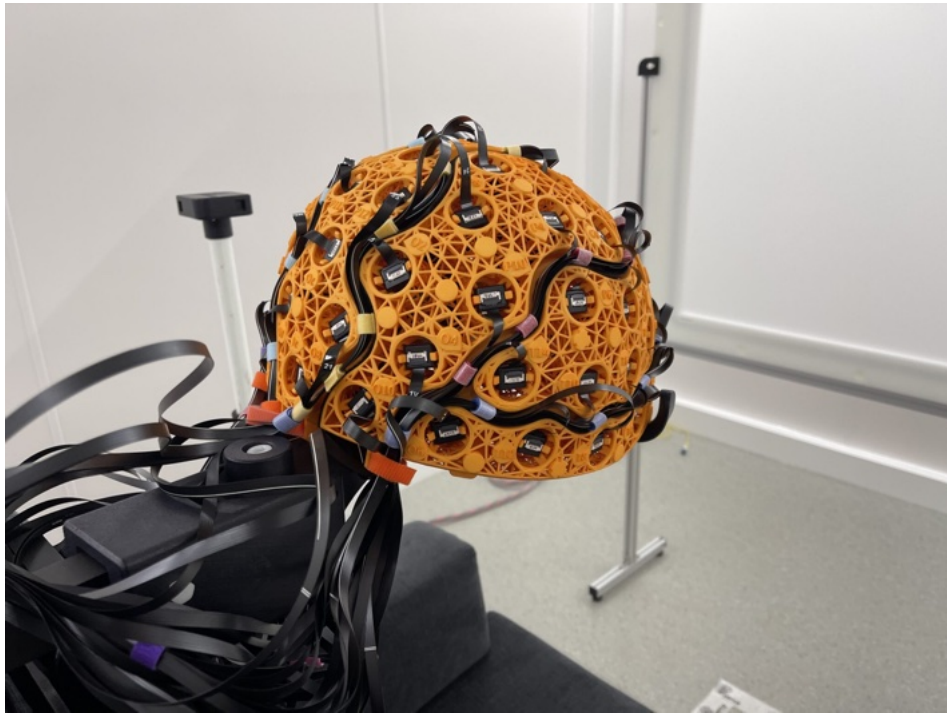
EEG: electroencephalography



fMRI: functional magnetic resonance imaging



MEG: magnetoencephalography



<https://scullycenter.princeton.edu/meg-lab/>

LENA: Language Environment Analysis

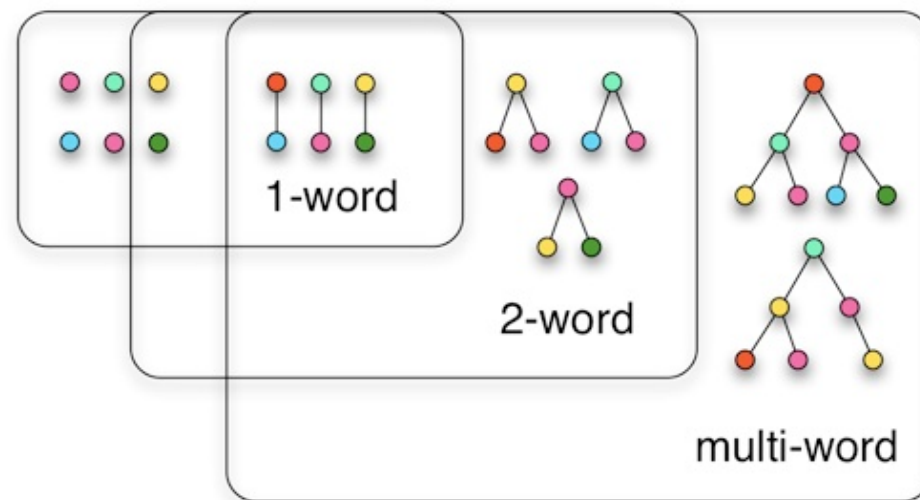
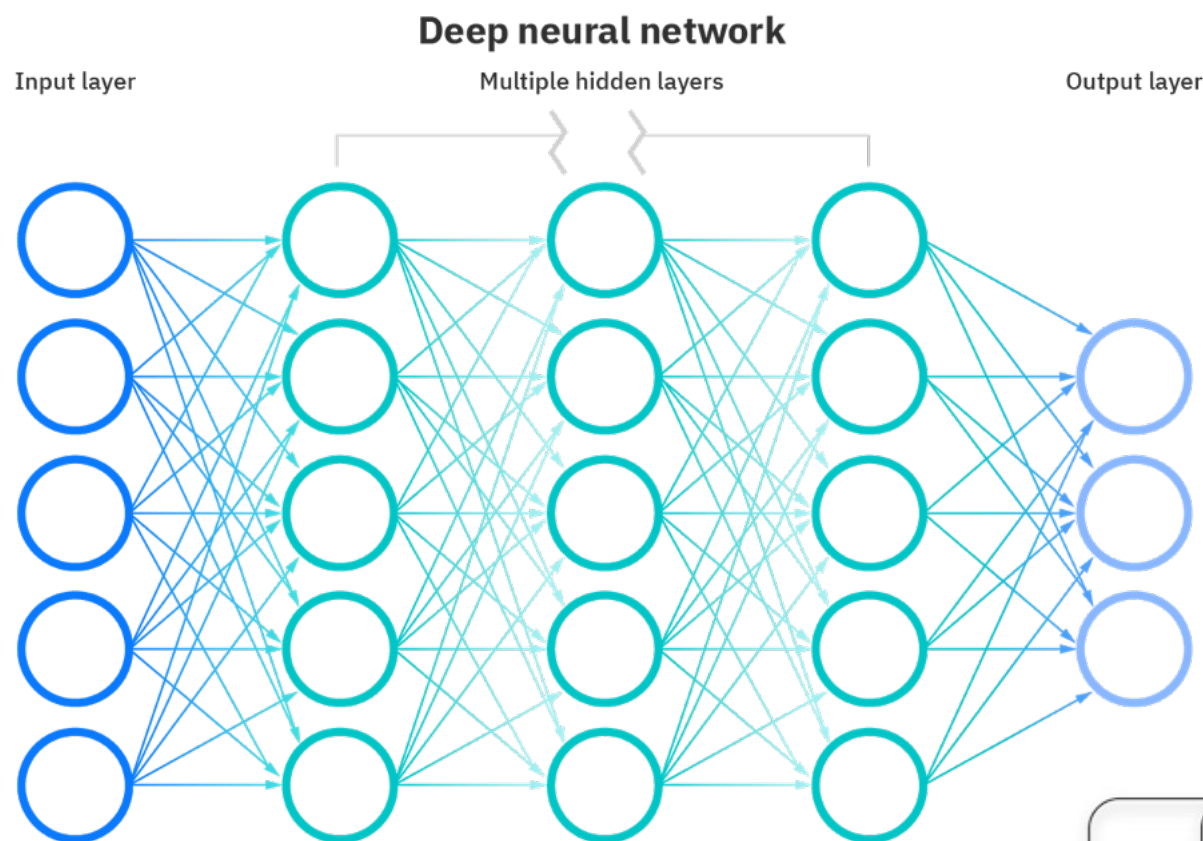


Records 16 hours of conversation
Automated counts of words and conversational turns

Touchscreens/tablets



Computational methods



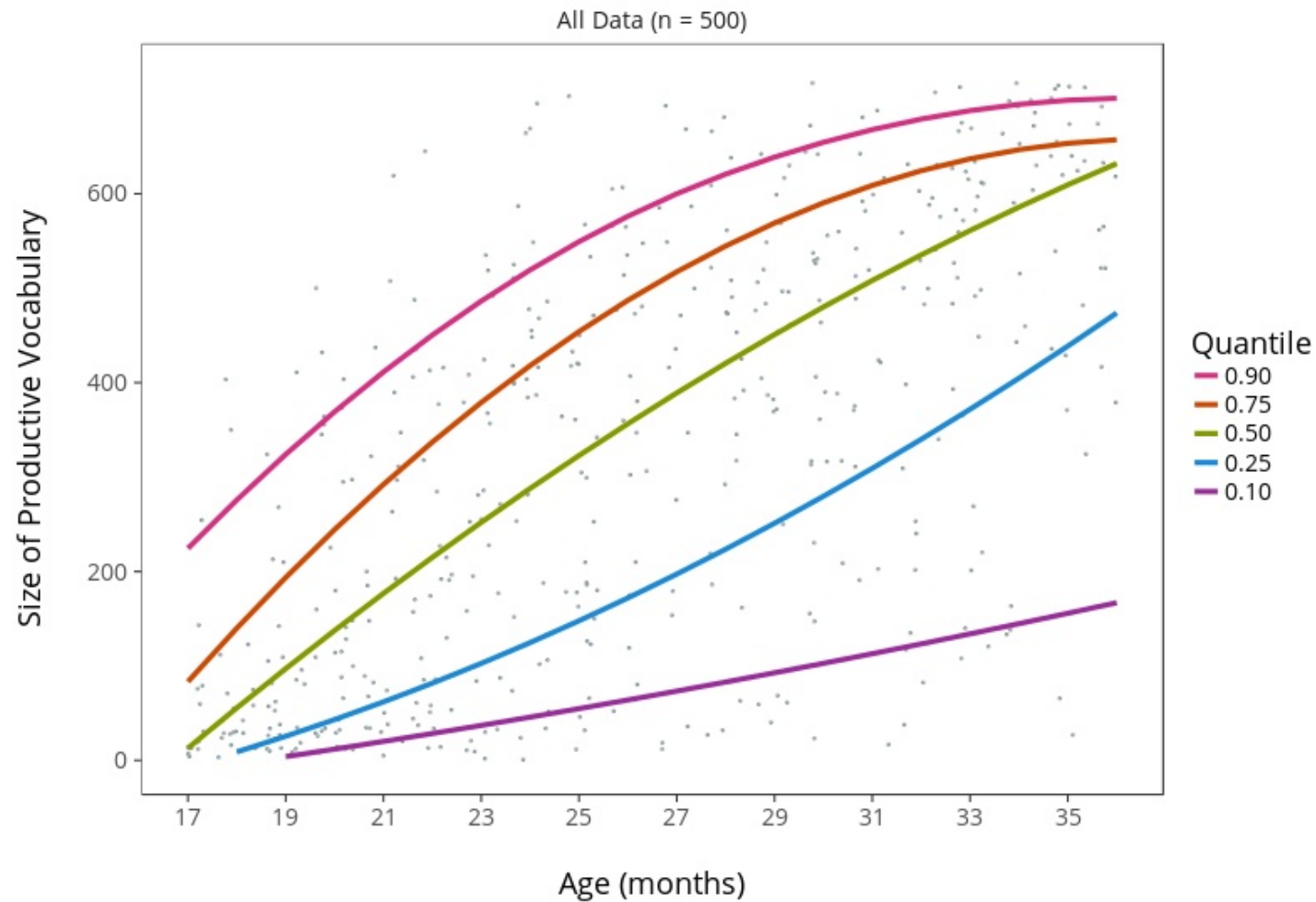
Analyses of parent-child speech

Child Language Data Exchange System (CHILDES)

22 *CHI: I xxx. ▶
23 *MOM: what you put in? ▶
24 *CHI: in that. ▶
25 *MOM: oh. ▶
26 *CHI: I threw that in. ▶
27 *MOM: oh honey. ▶
28 *DAD: what did you throw in huh? ▶
29 *DAD: you threw it in! ▶
30 *DAD: that's right. ▶
31 *DAD: you threw the baby in! ▶
32 *DAD: that's right. ▶
33 *DAD: this has been xxx. ▶
34 *DAD: xxx it's your boat play museum. ▶
35 *MOM: okay good girl. ▶
36 *DAD: okay well +/. ▶
37 *CHI: xxx. ▶
38 *MOM: okay you wanna lay down with your blanket on? ▶
39 *CHI: no. ▶
40 *MOM: okay you just sit and play there for a little while then. ▶
41 *MOM: that'll be fine. ▶
42 *MOM: we'll see you later, have a good nap. ▶
43 *DAD: have a good nap sweetie . ▶
44 *CHI: I I sit there xxx. ▶
45 *MOM: that's alright, you can sit there . ▶

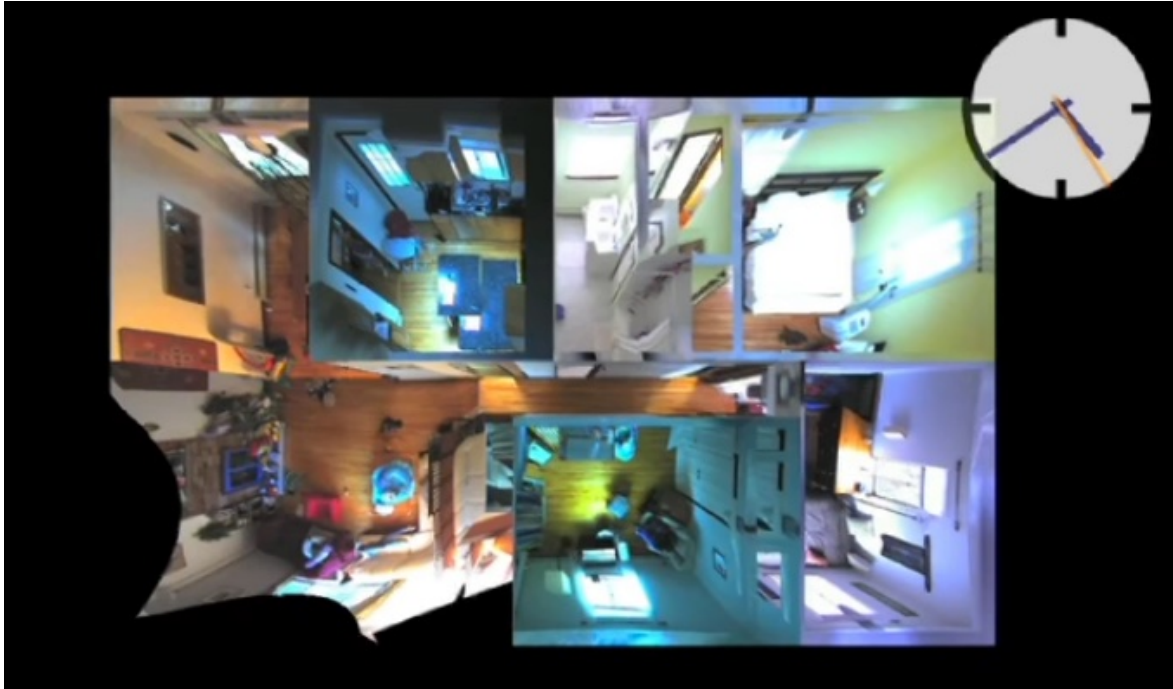
Big data: Surveys

WordBank



(This plot shows norms for Latvian-learning children)

Big data: natural home videos



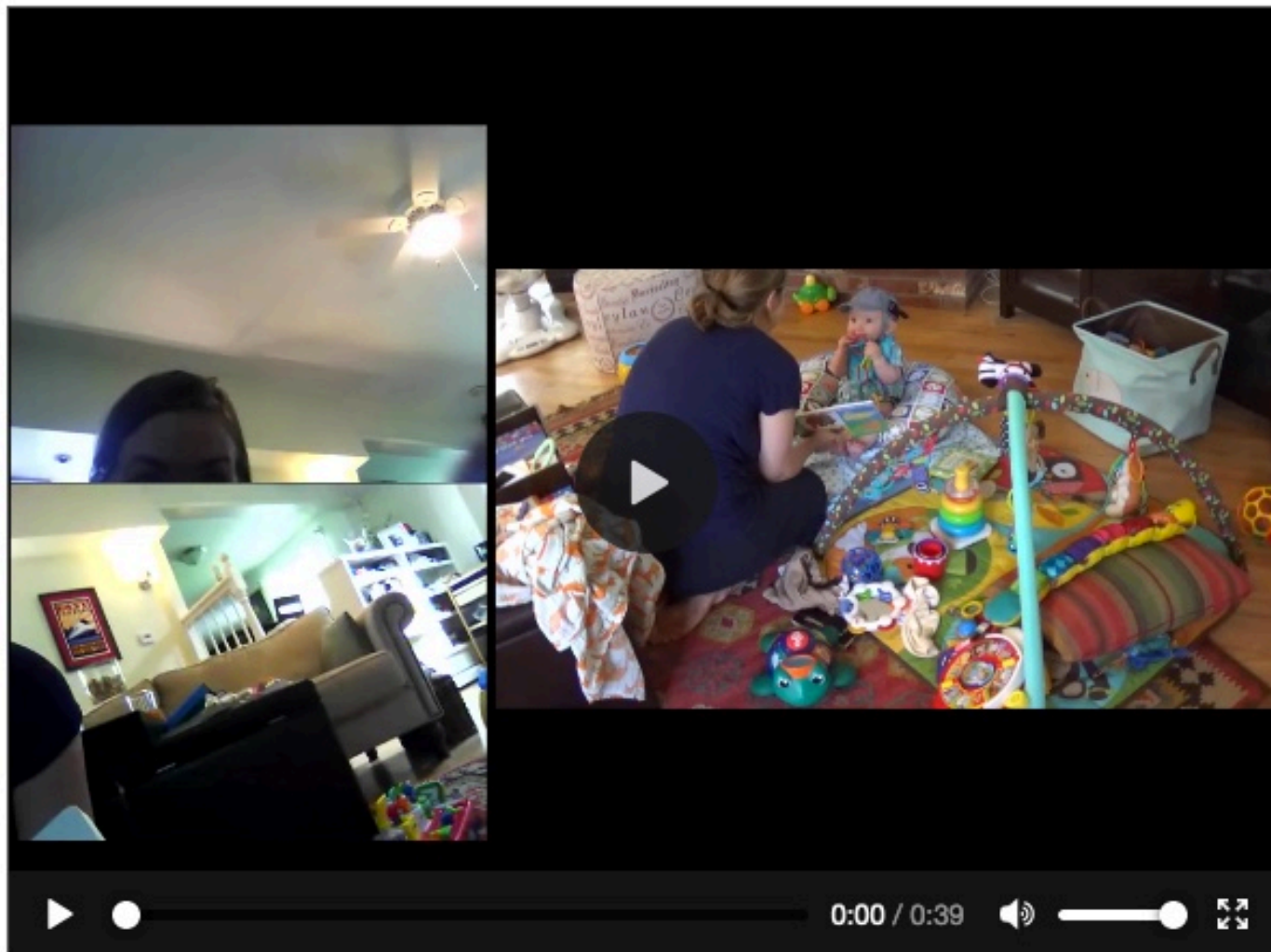
Big data: natural home videos





Databrary

Video data library for behavioral scientists



Developmental science requires
constant methodological innovation.
It's a deeply creative field.



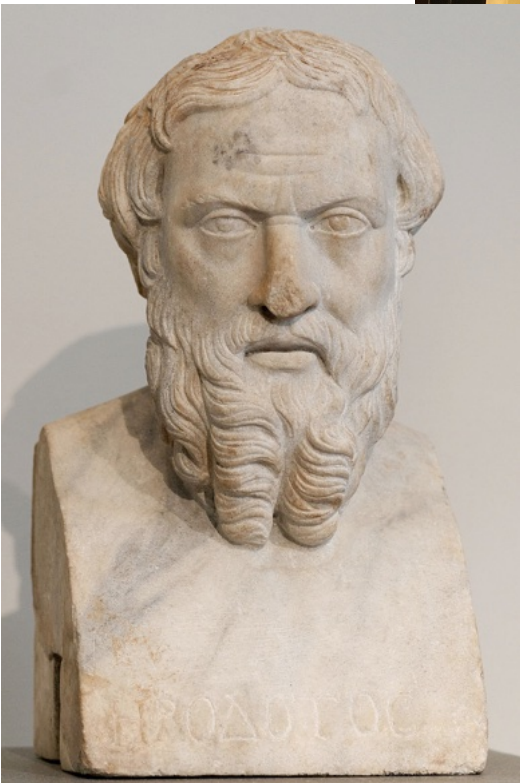
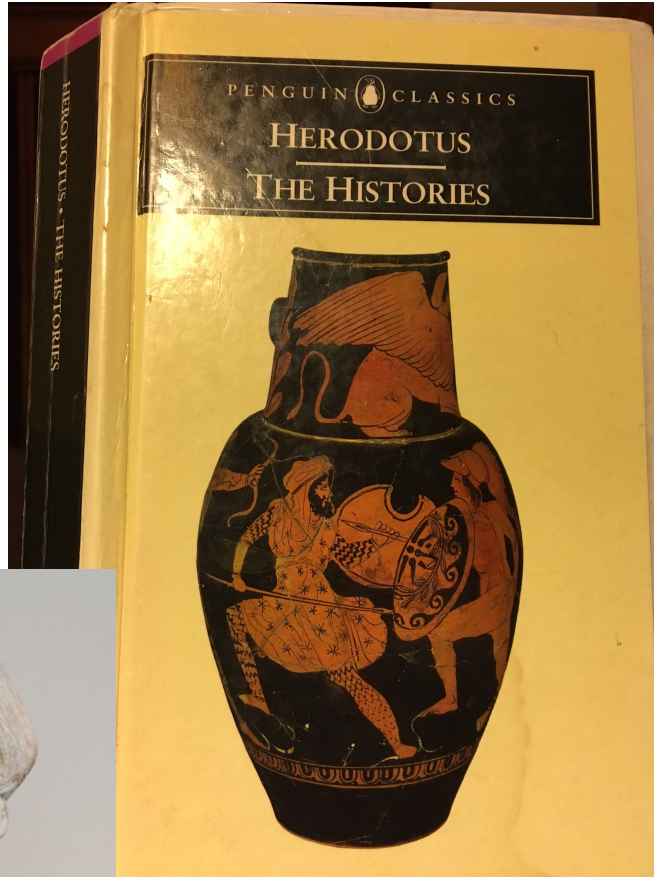
Keep in mind:
The scientific method
was invented by a philosopher.

René Descartes

June 8, 1637

*Discourse on the Method for Guiding
One's Reason and Searching for the
Truth in the Sciences*

5th-century BCE developmental methods



The Egyptians before the reign of Psammetichus used to think that of all races in the world they were the most ancient; Psammetichus, however, when he came to the throne, took it into his head to settle this question of priority, and ever since his time the Egyptians have believed that the Phrygians surpass them in antiquity and that they themselves come second.² Psammetichus, finding that mere inquiry failed to reveal which was the original race of mankind, devised an ingenious method of determining the matter. He took at random, from an ordinary family, two newly born infants and gave them to a shepherd to be brought up amongst his flocks, under strict orders that no one should utter a word in their presence. They were to be kept by themselves in a lonely cottage, and the shepherd was to bring in goats from time to time, to see that the babies had enough milk to drink, and to look after them in any other way that was necessary. All these arrangements were made by Psammetichus because he wished to find out what word the children would first utter, once they had grown out of their meaningless baby-talk. The plan succeeded; two years later the shepherd, who during that time had done everything he had been told to do, happened one day to open the door of the cottage and go in, when both children, running up to him with hands outstretched, pronounced the word 'becos'. The first time this occurred the shepherd made no mention of it; but later, when he found that every time he visited the children

to attend to their needs the same word was constantly repeated by them, he informed his master. Psammetichus ordered the children to be brought to him, and when he himself heard them say 'becos' he determined to find out to what language the word belonged. His inquiries revealed that it was the Phrygian for 'bread', and in consideration of this the Egyptians yielded their claims and admitted the superior antiquity of the Phrygians. That this was what really happened I myself learnt from the priests of Hephaestus at Memphis,³ though the Greeks have various improbable versions of the story, such as that Psammetichus had the children brought up by women whose tongues he had cut out. The version of the priests, however, is the one I have given. There were other things, too, which I learnt

